

Computer Architecture

CSIT — 3rd Semester — 2082 — Answer Sheet
Subject Code: CSC165

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain pipelining processing with an example of your own. Illustrate pipelining speed up equations under various conditions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Differentiate between restoring and non-restoring division algorithm. Explain the restoring division algorithm with an example of your own.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Describe the interrupt cycle. Explain the interrupt cycle with the required flowchart.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. List different arithmetic microorganisms. Explain 4-bit binary adder-subtractor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Briefly explain memory reference instructions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. List different binary codes. Explain any two of the binary codes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain any five addressing modes with an example of each.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define vector operation. Explain the concept of vector operation with an example of matrix multiplication.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Illustrate the working mechanism of Direct Memory Access in association with the concept of cycle stealing. How DMA is different from IOP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain microprogram sequencer for control memory with the required figure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Describe temporal locality and spatial locality. Differentiate between associative mapping and set associative mapping in cache memory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a) Handshaking b) Control word

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.